Business Questions – Open Payments Analysis

Oct 21, 2026 · @Osama

Summary

The data covers 2025 only: 16,131,856 payments worth $3.92 billion. Each question below is answered with plain SQL on the gold tables (fact_payment joined to its dimensions), run with answers.py.

- **Top manufacturer to physicians:** AbbVie Inc., $159.3M.
- **Top physician:** Andrew Cheng, $55.1M.
- **Most common payment:** Food and Beverage, 14.76M payments (91.5% of all). By amount, Royalty or License is largest at $1.20B (30.7%).
- **Top state:** Pennsylvania, $713.3M. **Top specialty:** Orthopaedic Surgery, $332.1M. **Top product:** Arthrex, $69.6M.
- **Trends (Q5, Q8):** only one year is loaded, so no increase or decrease can be read yet.

Q1. Which manufacturers are making the highest payments to physicians?

AbbVie Inc. pays physicians the most, $159.3M, ahead of Stryker at $119.9M. KPI: total payment amount by manufacturer.

```
SELECT m.manufacturer_name,
       ROUND(SUM(f.total_amount_usd), 2) AS total_paid
FROM gold.fact_payment f
JOIN gold.dim_manufacturer m ON m.manufacturer_sk = f.manufacturer_sk
JOIN gold.dim_recipient r    ON r.recipient_sk    = f.recipient_sk
WHERE r.recipient_type = 'Covered Recipient Physician'
GROUP BY m.manufacturer_name
ORDER BY total_paid DESC
LIMIT 10;
```

Manufacturer	Total paid (USD)
ABBVIE INC.	159,327,431.47
Stryker Corporation	119,911,317.20
Arthrex, Inc.	105,086,786.85
Medtronic, Inc.	102,565,997.55
Boston Scientific Corporation	85,941,513.80
Zimmer Biomet Holdings, Inc.	78,160,342.68
DePuy Synthes Products, Inc.	61,500,181.78
Novo Nordisk AS	61,355,768.33
INTUITIVE SURGICAL, INC.	56,404,319.68
Smith+Nephew, Inc.	48,699,391.21

Q2. Which physicians received the highest total payments?

Andrew Cheng received the most, $55.1M, almost double the second physician, Nitin Goyal at $30.0M. KPI: total payment amount by physician.

```
SELECT r.profile_id,
       r.first_name,
       r.last_name,
       ROUND(SUM(f.total_amount_usd), 2) AS total_received
FROM gold.fact_payment f
JOIN gold.dim_recipient r ON r.recipient_sk = f.recipient_sk
WHERE r.recipient_type = 'Covered Recipient Physician'
GROUP BY r.profile_id, r.first_name, r.last_name
ORDER BY total_received DESC
LIMIT 10;
```

Profile ID	First name	Last name	Total received (USD)
7227103	ANDREW	CHENG	55,070,157.30
372819	NITIN	GOYAL	30,000,067.88

Profile ID	First name	Last name	Total received (USD)
658915	MATTHEW	ALLAWAY	20,624,190.33
311622	KEVIN	FOLEY	20,179,978.53
105583	IVAN	OSORIO	18,183,201.14
1203562	RAMIN	FARZANEH-FAR	17,063,429.22
246552	FRED	LEE	14,065,875.66
439487	JONATHAN	EPSTEIN	13,127,853.53
755610	DREW	WEISSMAN	12,833,455.92
3539205	DAVID	DOVE	12,257,628.69

Q3. What type of payment is most common?

Food and Beverage is by far the most common: 14,764,648 payments, 91.5% of all, but only $443.4M. Royalty or License is the largest by amount, $1.20B from 15,496 payments. KPI: payment count and amount by nature of payment.

```
SELECT pt.nature_of_payment,
       COUNT(*)                          AS payment_count,
       ROUND(SUM(f.total_amount_usd), 2) AS total_amount
FROM gold.fact_payment f
JOIN gold.dim_payment_type pt ON pt.payment_type_sk = f.payment_type_sk
GROUP BY pt.nature_of_payment
ORDER BY payment_count DESC;
```

Nature of payment	Payment count	Total amount (USD)
Food and Beverage	14,764,648	443,396,829.52
Travel and Lodging	623,496	209,661,392.71
Compensation for services other than consulting, including serving as faculty or as a speaker at a venue other than a continuing education program	249,154	751,881,628.25
Consulting Fee	205,079	648,985,307.63

Nature of payment	Payment count	Total amount (USD)
Education	161,500	69,937,211.90
Gift	26,637	5,764,206.40
Honoraria	25,134	80,489,846.65
Long term medical supply or device loan	16,837	38,581,138.32
Royalty or License	15,496	1,203,443,512.93
Compensation for serving as faculty or as a speaker for a medical education program	15,039	34,131,280.73
Space rental or facility fees (teaching hospital only)	8,371	37,314,257.73
Grant	7,352	121,704,360.82
Debt forgiveness	6,179	48,590,175.68
Entertainment	5,849	506,793.38
Charitable Contribution	680	5,282,163.68
Acquisitions	405	223,880,856.47

Q4. Which states or cities have the highest payment activity?

Pennsylvania leads by amount at $713.3M, while Texas has the most payments (1,518,471). Philadelphia is the top city once its two spellings, PHILA and PHILADELPHIA, are added together: $600.2M. KPI: total payments by state and city.

By state:

```
SELECT r.state,
       COUNT(*)                          AS payment_count,
       ROUND(SUM(f.total_amount_usd), 2) AS total_amount
FROM gold.fact_payment f
JOIN gold.dim_recipient r ON r.recipient_sk = f.recipient_sk
WHERE r.state IS NOT NULL
GROUP BY r.state
ORDER BY total_amount DESC
LIMIT 10;
```

State	Payment count	Total amount (USD)
PA	689,644	713,306,030.33
CA	1,450,114	423,777,654.79
NY	1,089,903	260,029,754.53
TX	1,518,471	241,193,868.70
FL	1,436,414	231,575,568.52
MA	230,757	185,206,025.36
IL	535,247	126,470,447.98
OH	631,547	125,992,975.29
MO	331,289	125,550,488.11
MI	497,356	109,588,940.77

By city:

```
SELECT r.city,
       r.state,
       COUNT(*)                          AS payment_count,
       ROUND(SUM(f.total_amount_usd), 2) AS total_amount
FROM gold.fact_payment f
JOIN gold.dim_recipient r ON r.recipient_sk = f.recipient_sk
WHERE r.city IS NOT NULL
GROUP BY r.city, r.state
ORDER BY total_amount DESC
LIMIT 10;
```

City	State	Payment count	Total amount (USD)
PHILA	PA	1,790	537,279,217.23
NEW YORK	NY	200,667	107,903,620.52
Boston	MA	7,683	103,057,520.63
JEFFERSON CITY	MO	4,485	65,579,045.38

City	State	Payment count	Total amount (USD)
PHILADELPHIA	PA	91,269	62,921,266.00
SOUTH SAN FRANCISCO	CA	512	55,105,482.49
BOSTON	MA	48,200	51,114,863.54
DALLAS	TX	116,494	47,461,529.40
LOS ANGELES	CA	98,099	45,852,124.40
CHICAGO	IL	111,090	42,937,049.60

Q5. What is the trend of payments over time?

All 16,131,856 payments fall in 2025, so there is no year-over-year trend yet. Within 2025, April is the peak month at $655.5M and January the lowest at $212.1M. KPI: monthly, quarterly and yearly payment trend.

Yearly:

```
SELECT EXTRACT(year FROM f.date_of_payment) AS year,
       COUNT(*)                             AS payment_count,
       ROUND(SUM(f.total_amount_usd), 2)    AS total_amount
FROM gold.fact_payment f
WHERE f.date_of_payment IS NOT NULL
GROUP BY year
ORDER BY year;
```

Year	Payment count	Total amount (USD)
2025	16,131,856	3,923,550,962.80

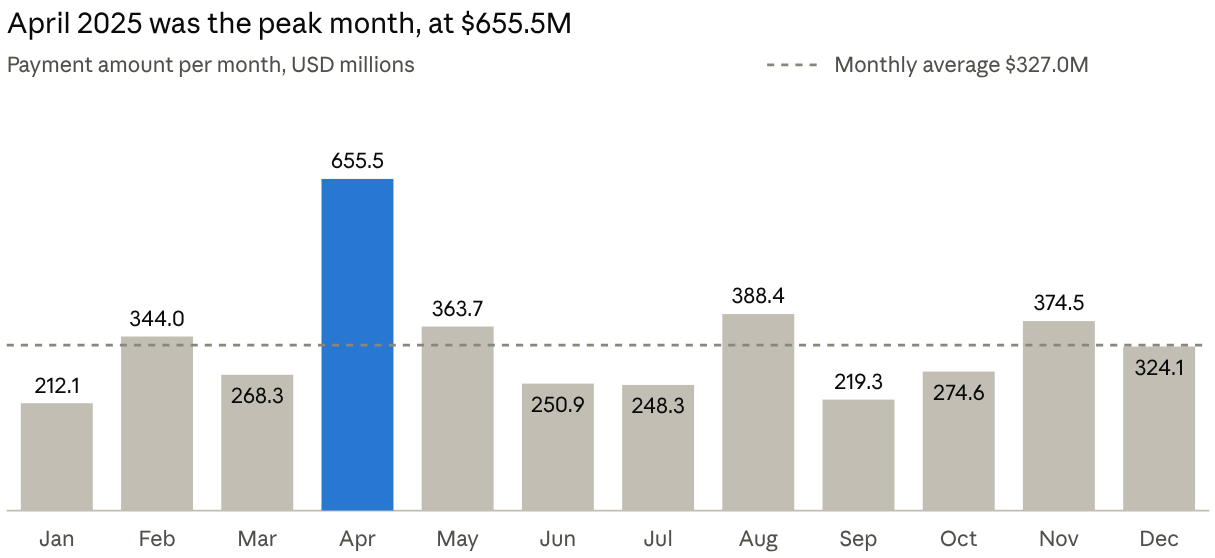
Quarterly:

```
SELECT EXTRACT(year FROM f.date_of_payment)    AS year,
       EXTRACT(quarter FROM f.date_of_payment) AS quarter,
       COUNT(*)                                AS payment_count,
       ROUND(SUM(f.total_amount_usd), 2)       AS total_amount
FROM gold.fact_payment f
WHERE f.date_of_payment IS NOT NULL
GROUP BY year, quarter
ORDER BY year, quarter;
```

Year	Quarter	Payment count	Total amount (USD)
2025	1	3,874,875	824,297,212.08
2025	2	4,250,727	1,270,062,566.31
2025	3	4,111,532	855,941,842.90
2025	4	3,894,722	973,249,341.51

Monthly:

```
SELECT EXTRACT(year FROM f.date_of_payment)  AS year,
       EXTRACT(month FROM f.date_of_payment) AS month,
       COUNT(*)                              AS payment_count,
       ROUND(SUM(f.total_amount_usd), 2)     AS total_amount
FROM gold.fact_payment f
WHERE f.date_of_payment IS NOT NULL
GROUP BY year, month
ORDER BY year, month;
```



answers.py output, Q5 monthly query · 2025, 12 months

Q6. Which medical specialties receive the highest payments?

Orthopaedic Surgery receives the most, $332.1M, well ahead of Internal Medicine at $193.2M. Sub-specialties are listed separately, for example Cardiovascular Disease under Internal Medicine. KPI: total payment amount by physician specialty.

```
SELECT r.specialty_1                     AS specialty,
       ROUND(SUM(f.total_amount_usd), 2) AS total_amount
FROM gold.fact_payment f
JOIN gold.dim_recipient r ON r.recipient_sk = f.recipient_sk
WHERE r.specialty_1 IS NOT NULL
GROUP BY r.specialty_1
ORDER BY total_amount DESC
LIMIT 10;
```

Specialty	Total amount (USD)
Allopathic & Osteopathic Physicians|Orthopaedic Surgery	332,135,135.40
Allopathic & Osteopathic Physicians|Internal Medicine	193,211,704.85
Allopathic & Osteopathic Physicians|Psychiatry & Neurology|Neurology	102,871,441.34

Specialty	Total amount (USD)
Allopathic & Osteopathic Physicians|Dermatology	101,838,789.86
Allopathic & Osteopathic Physicians|Neurological Surgery	94,156,041.13
Allopathic & Osteopathic Physicians|Orthopaedic Surgery|Adult Reconstructive Orthopaedic Surgery	86,603,187.96
Allopathic & Osteopathic Physicians|Internal Medicine|Cardiovascular Disease	85,537,989.84
Allopathic & Osteopathic Physicians|Psychiatry & Neurology|Psychiatry	80,740,181.29
Allopathic & Osteopathic Physicians|Internal Medicine|Hematology & Oncology	71,879,874.05
Allopathic & Osteopathic Physicians|Internal Medicine|Gastroenterology	71,010,082.61

Q7. Which drugs or medical products are linked with the highest payments?

Arthrex is linked with the most money, $69.6M, followed by the drug Pluvicto at $65.3M. A payment can name up to 5 products, so each amount is split between them using the bridge table's allocation weight, and nothing is counted twice. KPI: total payment amount by drug or product.

```
SELECT p.product_name,
       ROUND(SUM(f.total_amount_usd * b.allocation_weight), 2) AS
total_amount
FROM gold.fact_payment f
JOIN gold.bridge_payment_product b ON b.payment_sk = f.payment_sk
JOIN gold.dim_product p            ON p.product_sk = b.product_sk
WHERE p.product_name IS NOT NULL
GROUP BY p.product_name
ORDER BY total_amount DESC
LIMIT 10;
```

Product	Total amount (USD)
Arthrex	69,633,079.32
PLUVICTO	65,258,444.03
Da Vinci Surgical System	50,329,561.08
General - Therapies	47,372,402.78
DUPIXENT	43,919,670.37
Tecentriq	43,142,482.83
BOTOX	33,785,403.39
TREMFYA	28,861,494.23
ATTUNE	24,394,453.19
ACCOLADE	23,832,003.59

Q8. Which payment categories are increasing or decreasing over time?

No category can be called increasing or decreasing yet: every one has a single year, 2025. The query is ready, and once earlier years are loaded the same SQL will show each category's change. KPI: trend by payment category (nature of payment).

```
SELECT pt.nature_of_payment,
       EXTRACT(year FROM f.date_of_payment) AS year,
       ROUND(SUM(f.total_amount_usd), 2)    AS total_amount
FROM gold.fact_payment f
JOIN gold.dim_payment_type pt ON pt.payment_type_sk = f.payment_type_sk
WHERE f.date_of_payment IS NOT NULL
GROUP BY pt.nature_of_payment, year
ORDER BY pt.nature_of_payment, year;
```

Nature of payment	Year	Total amount (USD)
Acquisitions	2025	223,880,856.47
Charitable Contribution	2025	5,282,163.68

Nature of payment	Year	Total amount (USD)
Compensation for services other than consulting, including serving as faculty or as a speaker at a venue other than a continuing education program	2025	751,881,628.25
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